Galileo E5 AltBOC Implementation Notes

This note summarizes the current PocketSDR implementation for Galileo E5 wideband AltBOC tracking.

Scope

The implemented signal is `E5ABQ`.

`E5ABQ` is a pilot-only, sign-only AltBOC approximation that combines the Galileo E5aQ and E5bQ pilot components at the E5 center frequency:

- RF frequency: `1191.795 MHz`
- Code period: `1 ms`
- Base code length: `10230 chips`
- Internal complex replica length: `40920 samples/chips`
- RINEX observation code mapping: `CODE_L8Q`

The design intentionally does not implement ICD-perfect full E5 AltBOC with data-assisted product-term wipeoff.

Signal Model

The replica is built from the four AltBOC subcarriers defined in Galileo OS SIS ICD Issue 2.1, section 2.1.1 (E5 signal generation). The two I-channel subcarriers (lower / upper sideband) are stored as sign-only 8-PSK tables at 4 samples/chip:

- `SC_E5aI_I`, `SC_E5aI_Q`   (sc_E5aI = sc_S - j*sc_P, lower sideband)
- `SC_E5bI_I`, `SC_E5bI_Q`   (sc_E5bI = sc_S + j*sc_P, upper sideband)

The Q-channel subcarriers per ICD are 90 deg rotations of the corresponding I-channel ones, with **opposite signs** between E5a and E5b:

```
sc_E5aQ = +j * sc_E5aI
sc_E5bQ = -j * sc_E5bI
```

The ICD pilot Q-only signal is therefore

```
S_Q(t) = c_E5aQ * sec_E5aQ * sc_E5aQ + c_E5bQ * sec_E5bQ * sc_E5bQ
       = j * [ c_E5aQ * sec_E5aQ * sc_E5aI - c_E5bQ * sec_E5bQ * sc_E5bI ]
```

Letting `rel = sec_E5aQ * sec_E5bQ`, the matched local replica (after the global +j is absorbed by the Costas PLL) is

```
R(t) = c_E5aQ * sc_E5aI - rel * c_E5bQ * sc_E5bI
```

This is the bracketed expression generated by `mod_code_cpx2_periodic` in src/sdr_code.c. The combined I/Q values are sign-limited to `{-1, 0, +1}` so the existing 8-bit complex correlator path can be reused.

Acquisition

Acquisition uses the E5aQ AltBOC complex replica only. This avoids requiring secondary-code alignment before signal detection.

Once the signal is acquired, the channel starts tracking with the same E5aQ AltBOC replica until the E5aQ secondary code is synchronized.

Tracking

Tracking uses three precomputed code banks:

1. Bank 0: `c_E5aQ * sc_E5aI`, used before secondary-code sync.
2. Bank 1: `c_E5aQ * sc_E5aI - c_E5bQ * sc_E5bI` (matches `rel_actual = +1`).
3. Bank 2: `c_E5aQ * sc_E5aI + c_E5bQ * sc_E5bI` (matches `rel_actual = -1`).

At each 1 ms tracking epoch, if the E5aQ secondary code is synchronized, the receiver computes `rel = sec_E5aQ * sec_E5bQ` and selects bank 1 (`rel > 0`) or bank 2 (`rel < 0`).

The correlator uses a complex-code dot product:

```
corr = data * conj(code)
```

This is required because AltBOC replicas have distinct I and Q code values.

Navigation Decoding

`E5ABQ` is routed through the existing E5aQ pilot navigation path. It is used as a tracking and observation signal, not as an E5aI/E5bI data-channel decoder.

Data-channel demodulation remains the responsibility of the existing `E5AI` and `E5BI` channels.

Rejected Variants

`E5IQ`

An earlier Tier 2 signal name, `E5IQ`, represented E5aQ-only AltBOC tracking. It was useful for bring-up, but was removed from the public signal list after `E5ABQ` became stable.

The E5aQ-only AltBOC replica still exists internally as the acquisition and pre-secondary-sync fallback for `E5ABQ`.

E5ABQF

An experimental full AltBOC-Q signal, `E5ABQF`, was also tested. It included the E5 product terms and used four E5aI/E5bI data-sign hypotheses.

This was rejected because full AltBOC product terms require reliable wipeoff of the E5aI and E5bI navigation data signs. Without data-assisted wipeoff, the tracking loop sees unstable phase changes and split correlation peaks.

Supporting full AltBOC robustly would require sharing data-sign state from separate E5AI/E5BI channels or adding a dedicated data-assisted wideband tracking architecture. That is intentionally out of scope for the current implementation.

Expected Behavior

Compared with E5aQ or E5aI alone, `E5ABQ` can use both E5aQ and E5bQ pilot power. The ideal pilot-power gain is about `3 dB` over a single pilot component.

In practice, the measured gain can be smaller because of:

- frontend passband ripple and roll-off,
- E5a/E5b gain imbalance,
- sign-only replica approximation,
- receiver C/N0 estimator assumptions,
- finite front-end bandwidth.

The implementation should therefore be judged primarily by stable lock, consistent code offset versus E5a/E5b sideband channels, and a clean single correlation peak.